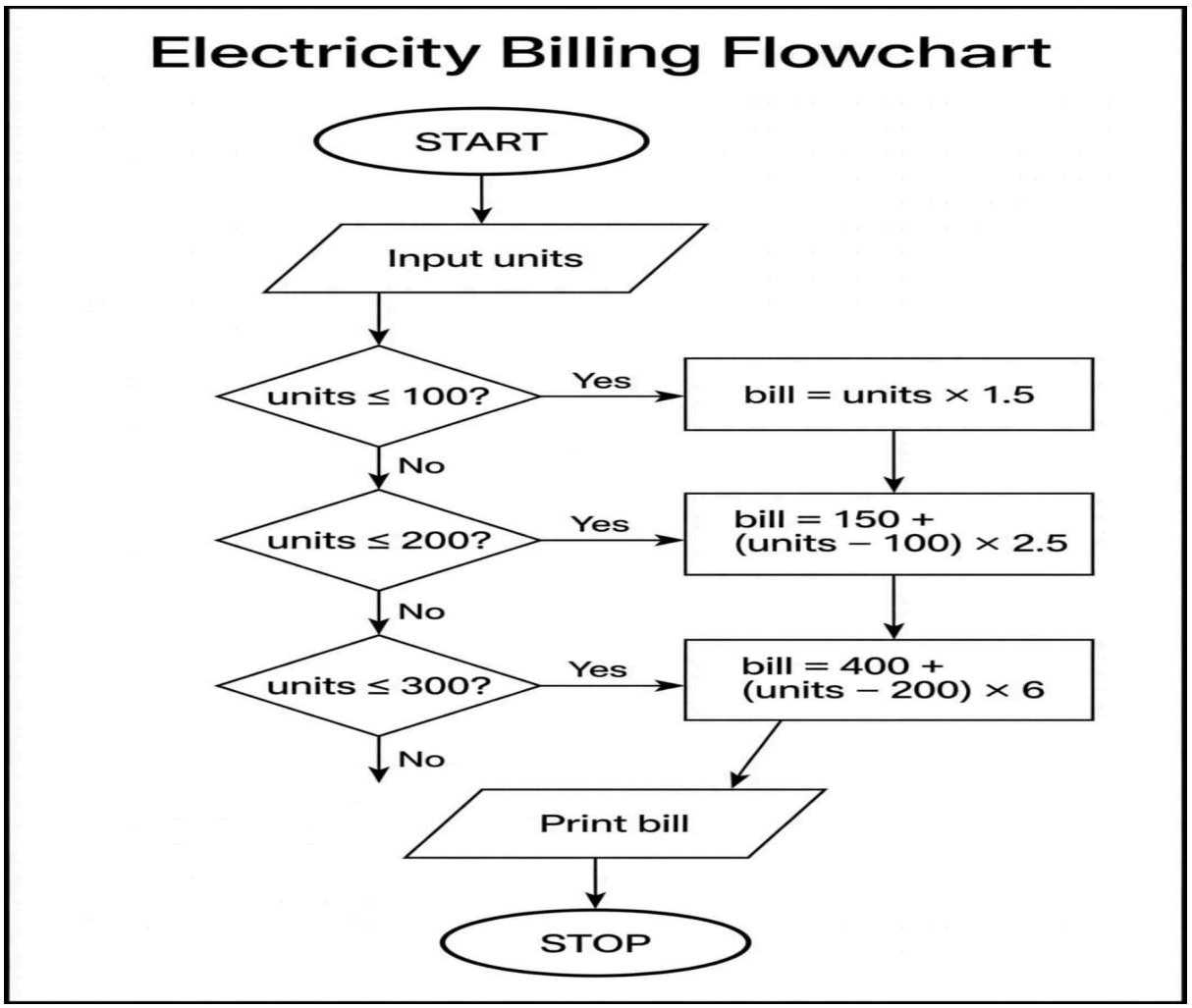
Problem Statement: Calculate electricity bill

Objective:We use if-elif condition in this program.

Algorithm for Electricity bill:

1.Start the program
2.Ask user for units
3.Check which slab the units fall
4.Calculate bill as per slab rate
5.Show bill amount
6.Stop

Flowchart:


Electricity Billing Flowchart
START
Input units
units ≤ 100?
Yes
bill = units × 1.5
No
units ≤ 200?
Yes
bill = 150 +
(units − 100) × 2.5
No
units ≤ 300?
Yes
bill = 400 +
(units − 200) × 6
No
Print bill
STOP

Code:

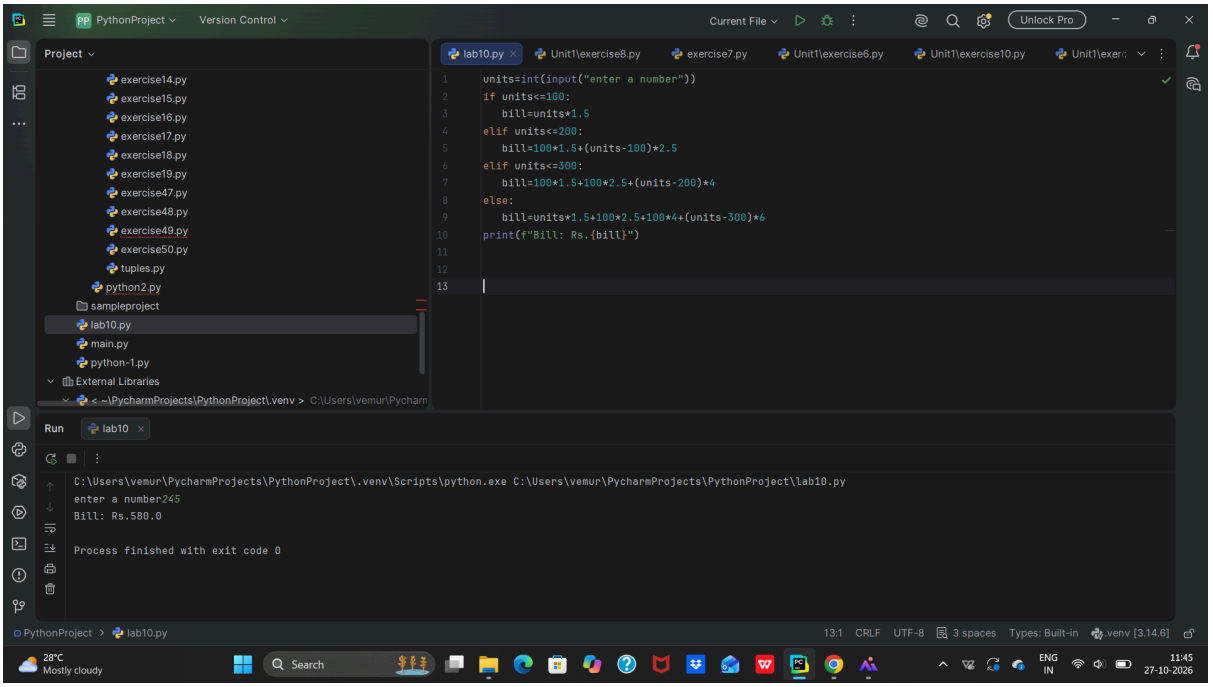
```
units=int(input("enter a number"))
if units<=100:
  bill=units*1.5
elif units<=200:
  bill=100*1.5+(units-100)*2.5
elif units<=300:
  bill=100*1.5+100*2.5+(units-200)*4
else:
  bill=units*1.5+100*2.5+100*4+(units-300)*6
print(f"Bill: Rs.{bill}")
```

Output:

Enter units: 234

Bill :Rs.536.0

Screenshot:



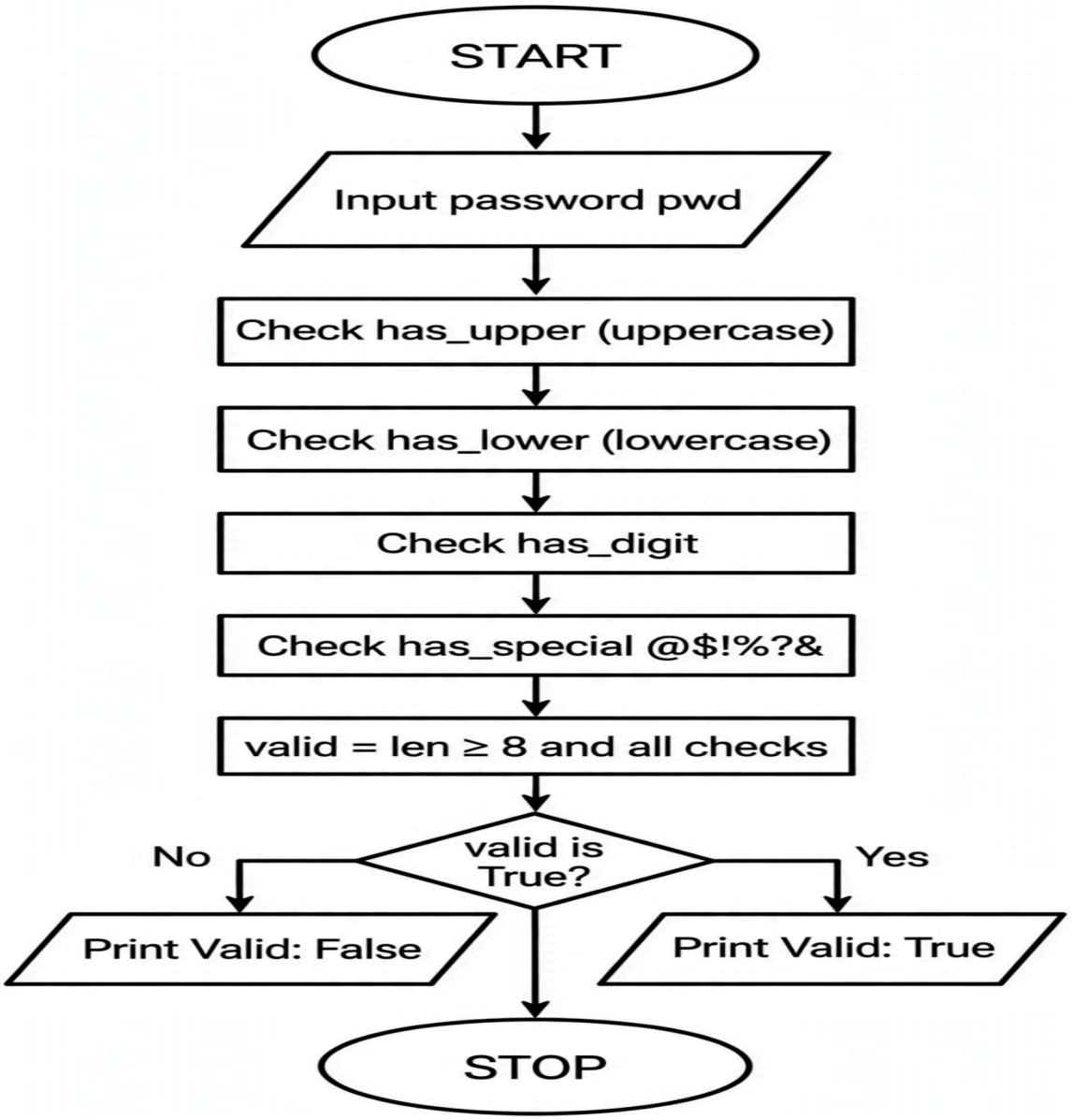
Problem Statement: Calculate password validation

Algorithm for  password validation:

1.Start
2.Take password from users
3.Check password has capital letters,small letters,numbers and symbols
4.Check password length is 8 or more
5.If all true - Valid
6.ELSE - Not Valid
7.Print result
8.Stop

Flowchart:

# Password Validation Flowchart

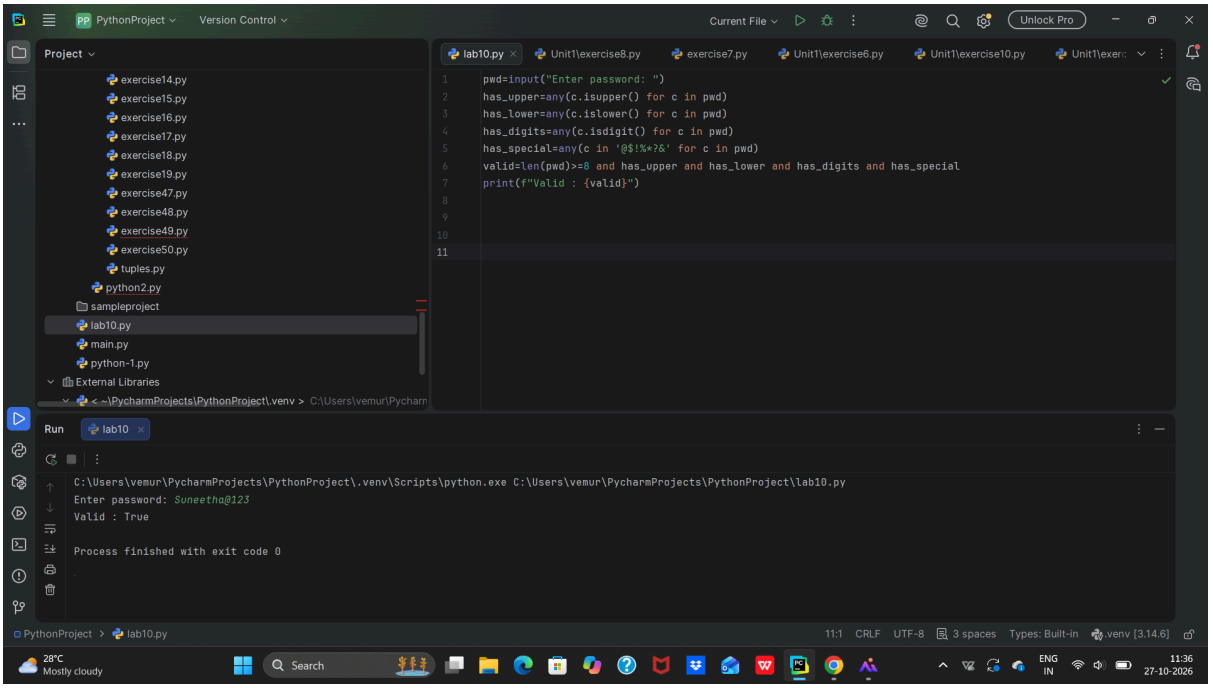


Code:

```
pwd=input("Enter password: ")
has_upper=any(c.isupper() for c in pwd)
has_lower=any(c.islower() for c in pwd)
has_digits=any(c.isdigit() for c in pwd)
has_special=any(c in '@$!%*?&' for c in pwd)
valid=len(pwd)>=8 and has_upper and has_lower and has_digits and has_special
print(f"Valid : {valid}")
```

Output:
Enter password: Suneetha@123
Valid : True

Screenshot:

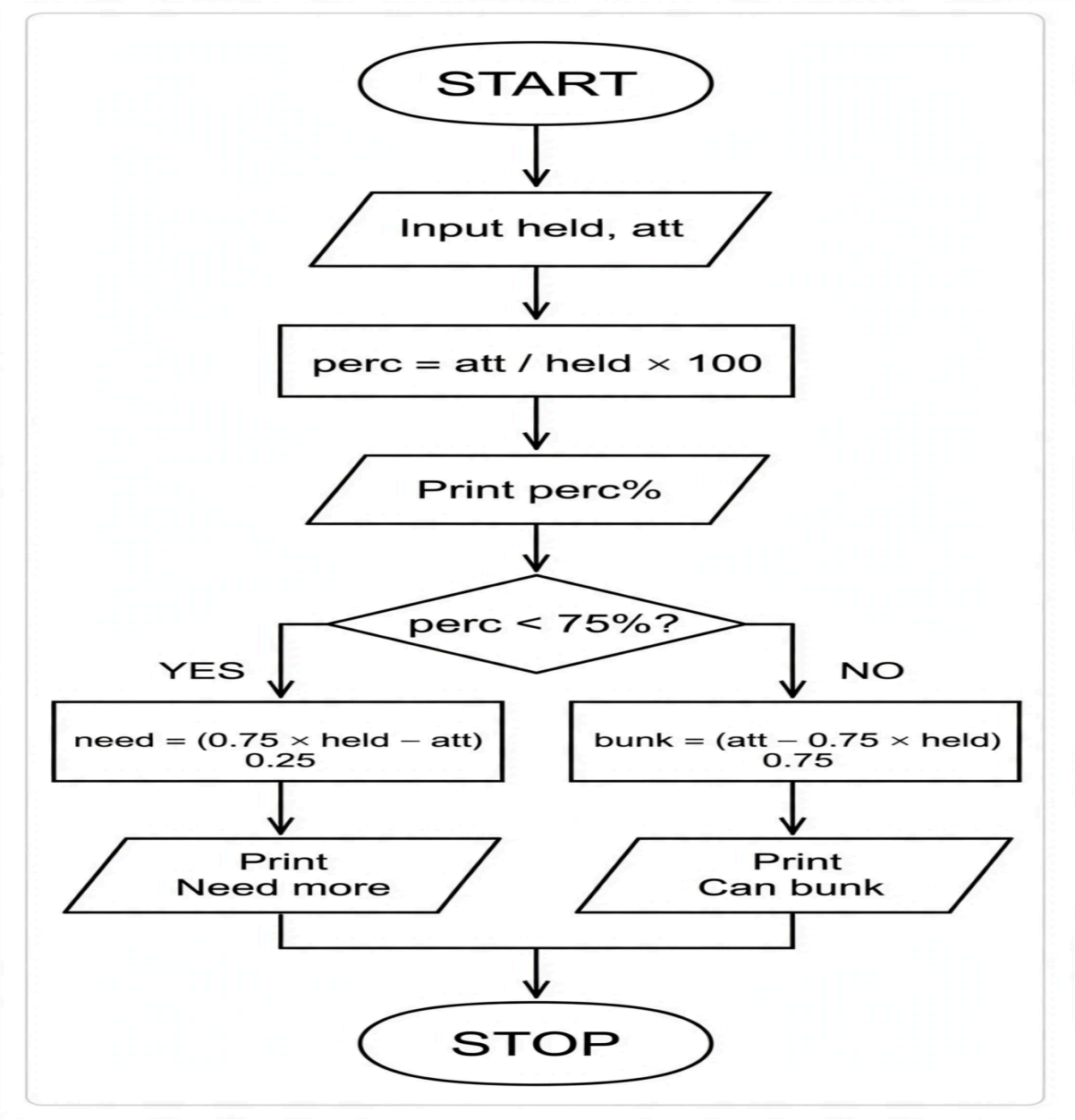


Problem Statement: Calculate Attendance recovery

Algorithm for attendance recovery:

1.Start
2.Enter held and attended
3.Find % = att/held*100
4.If % <75 show need,else show bunk
5.Stop

Flowchart:

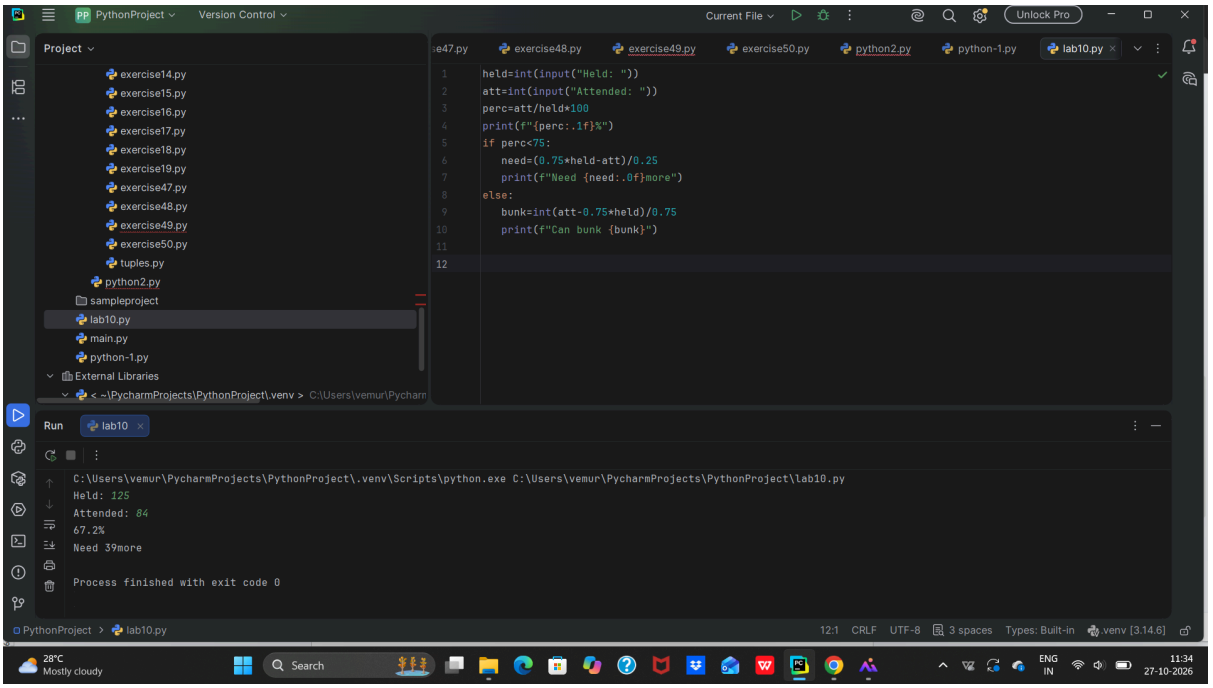


Code:

```
held=int(input("Held: "))
att=int(input("Attended: "))
perc=att/held*100
print(f"{perc:.1f}%")
if perc<75:
   need=(0.75*held-att)/0.25
   print(f"Need {need:.0f}more")
else:
   bunk=int(att-0.75*held)/0.75
   print(f"Can bunk {bunk}")
```

Output:
Held: 125
Attended: 84
67.2%
Need 39more

Screenshot:

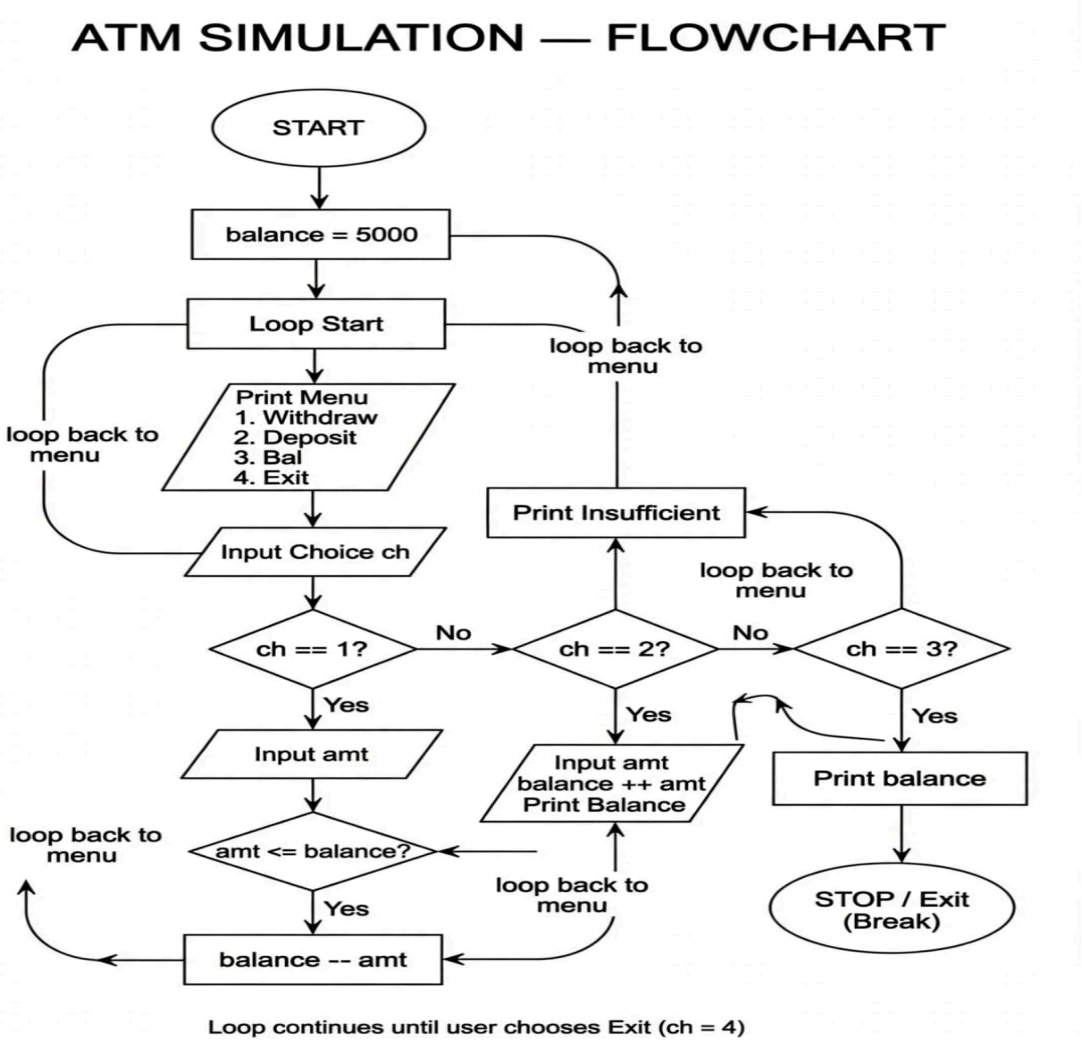


Problem Statement: Calculate ATM Simulation

Algorithm for atm simulation:

1.Start
2.balance=5000
3.Print menu and take choice
4.If choice=1 check balance and withdraw
5.If choice=2 add amount to balance
6.If choice=3 print balance
7.If choice=4 exit loop
8.Stop

Flowchart:




Code:

```
balance=5000
while True:
```

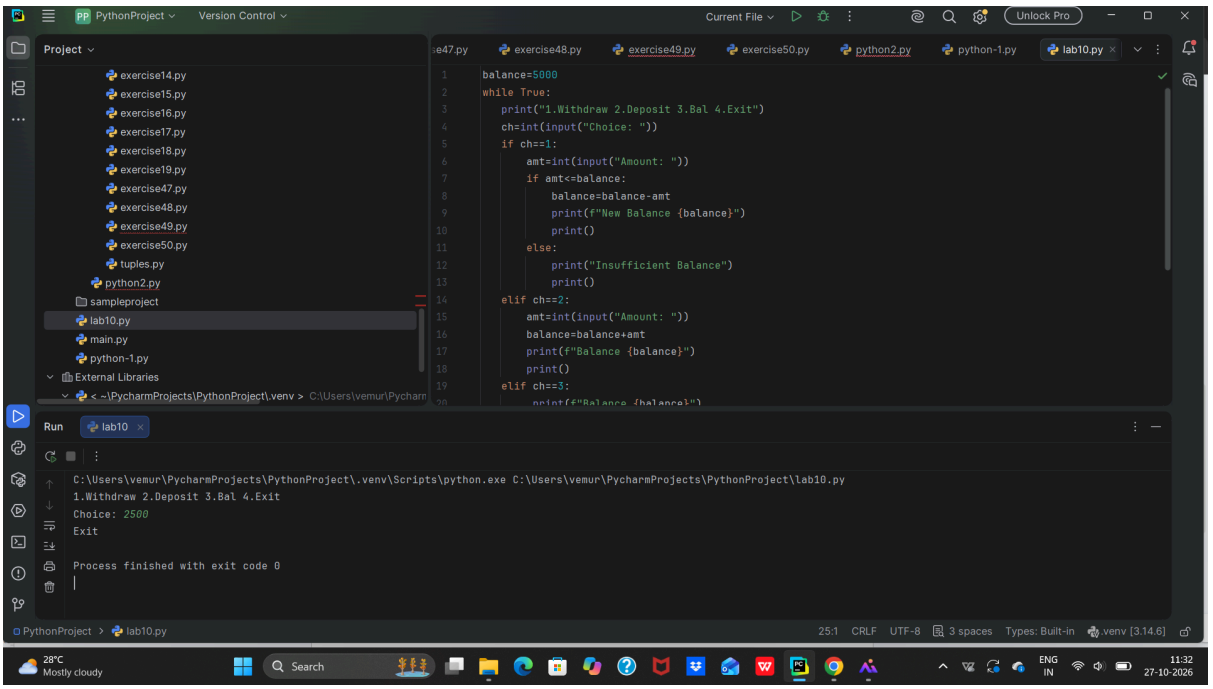
```
print("1.Withdraw 2.Deposit 3.Bal 4.Exit")
ch=int(input("Choice: "))
if ch==1:
    amt=int(input("Amount: "))
    if amt<=balance:
        balance=balance-amt
        print(f"New Balance {balance}")
        print()
    else:
        print("Insufficient Balance")
        print()
elif ch==2:
    amt=int(input("Amount: "))
    balance=balance+amt
    print(f"Balance {balance}")
    print()
elif ch==3:
     print(f"Balance {balance}")
     print()
else:
    print("Exit")
    break
```

Output:
1.Withdraw 2.Deposit 3.Bal 4.Exit
Choice: 1
Amount: 1000
New Balance 4000

1.Withdraw 2.Deposit 3.Bal 4.Exit
Choice: 2
Amount: 500
Balance 4500

1.Withdraw 2.Deposit 3.Bal 4.Exit
Choice: 4
Exit

Screenshot:


PythonProject
Version Control
Current File
Unlock Pro
Project
exercise14.py
exercise15.py
exercise16.py
exercise17.py
exercise18.py
exercise19.py
exercise47.py
exercise48.py
exercise49.py
exercise50.py
tuples.py
python2.py
sampleproject
lab10.py
main.py
python-1.py
External Libraries
exercise48.py
exercise49.py
exercise50.py
python2.py
python-1.py
lab10.py
balance=5000
while True:
print("1.Withdraw 2.Deposit 3.Bal 4.Exit")
ch=int(input("Choice: "))
if ch==1:
amt=int(input("Amount: "))
if amt<=balance:
balance=balance-amt
print(f"New Balance {balance}")
print()
else:
print("Insufficient Balance")
print()
elif ch==2:
amt=int(input("Amount: "))
balance=balance+amt
print(f"Balance {balance}")
print()
elif ch==3:
Run
lab10
C:\Users\vemur\PycharmProjects\PythonProject\.venv\Scripts\python.exe C:\Users\vemur\PycharmProjects\PythonProject\lab10.py
1.Withdraw 2.Deposit 3.Bal 4.Exit
Choice: 2500
Exit
Process finished with exit code 0
PythonProject
lab10.py
25:1
CRLF
UTF-8
3 spaces
Types: Built-in
venv (3.14.6)
28°C
Mostly cloudy
Search
ENG
IN
11:32
27-10-2026

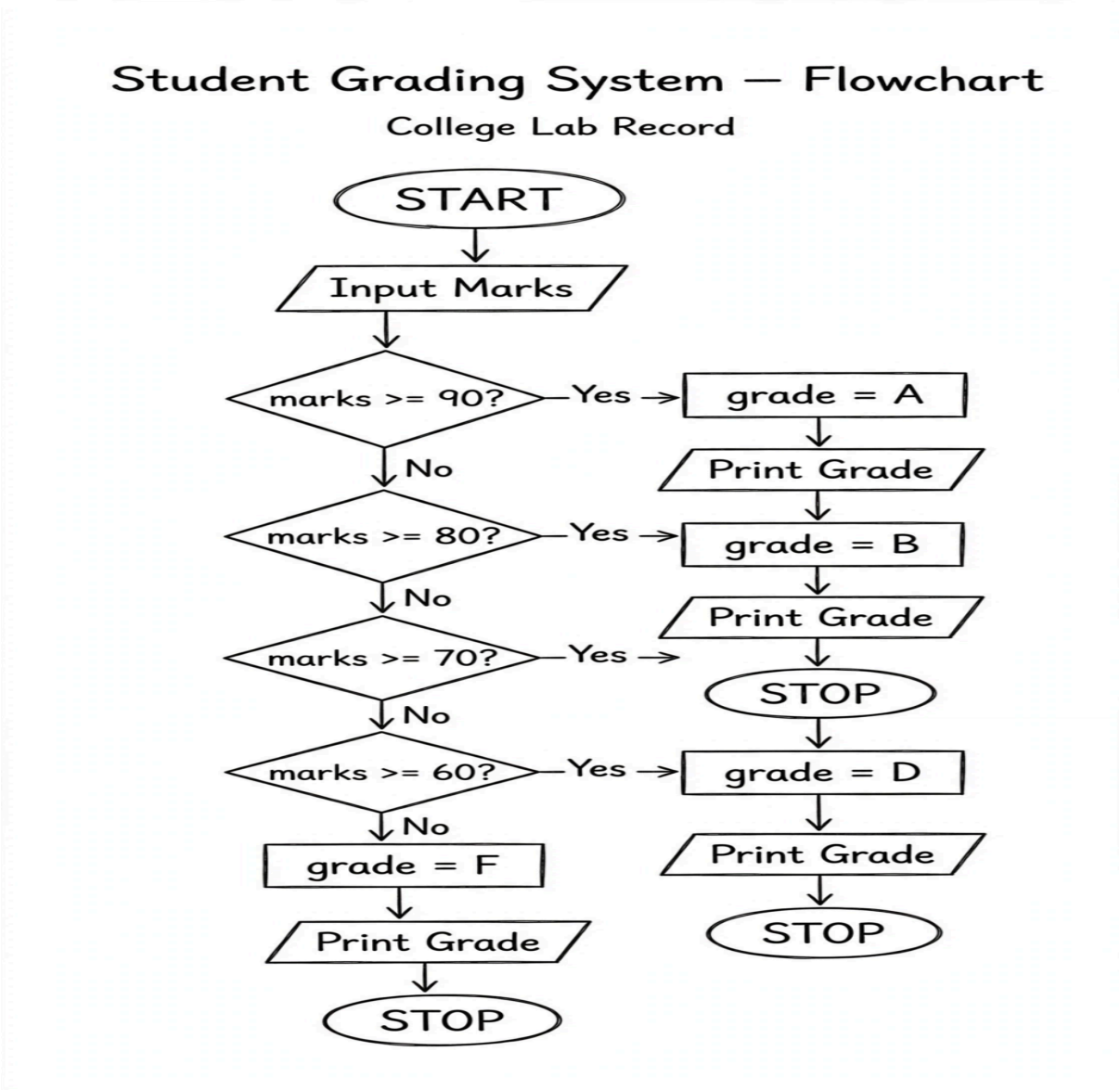
Problem Statement: Calculate student grading

Objective:We use if-elif condition in this program.

Algorithm for student grading:

1.Start the program
2.Input marks
3.Check marks by using if-elif ladder
4.Print grade
5.Stop

Flowchart:




Code:

```
marks=int(input())
if marks>=90:
  grade="A"
elif marks>=80:
```

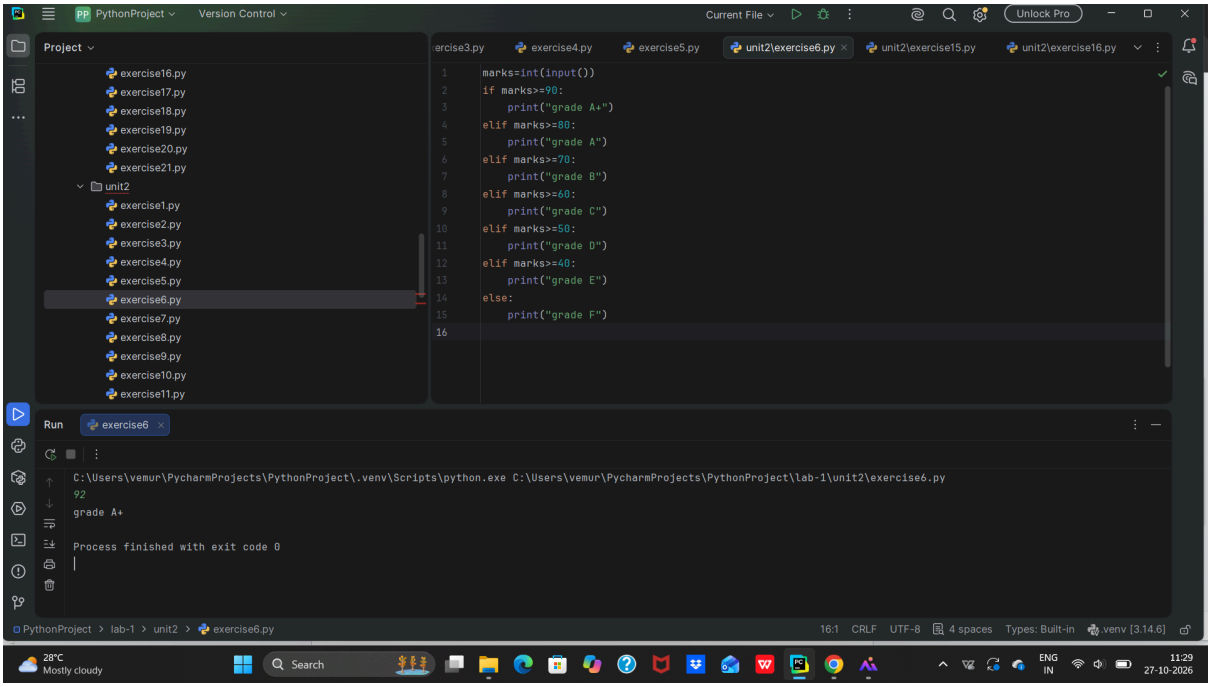
```
  grade="B"
elif marks>=70:
    grade="C"
elif marks>=60:
    grade="D"
else:
    grade="F"
print(f"Grade:{grade}")
```

Output:

85

Grade:B

Screenshot:

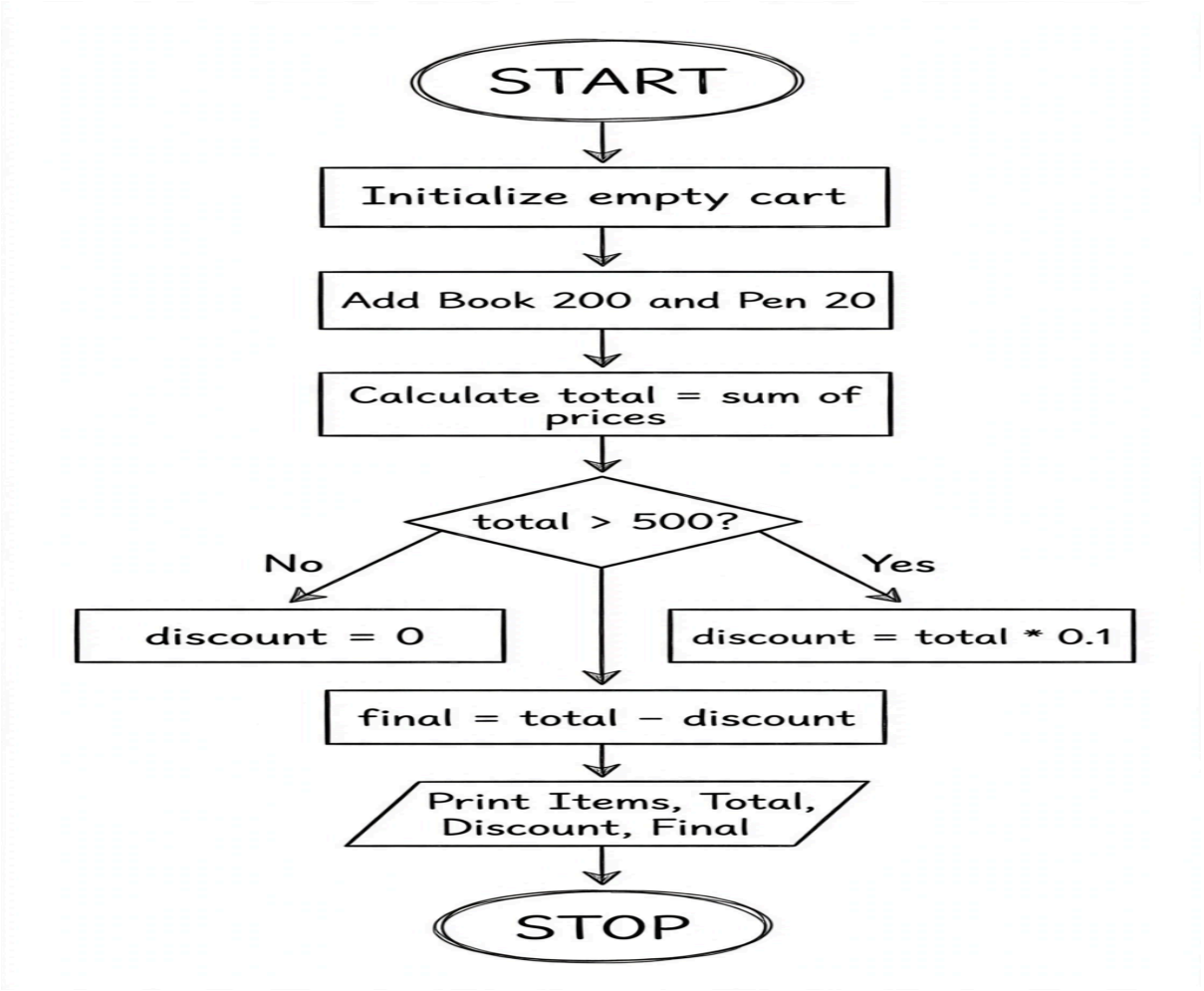


Problem Statement: Calculate shopping cart

Algorithm for shopping cart:

1.Start
2.Create empty cart[ ]
3.Add Books=200 and Pen=20 to cart
4.Total =sum of all prices
5.If total>500 THEN discount=total*0.1 ELSE discount=0
6.final = total-discount
7.Print cart,total,discount,final
8.Stop

Flowchart:



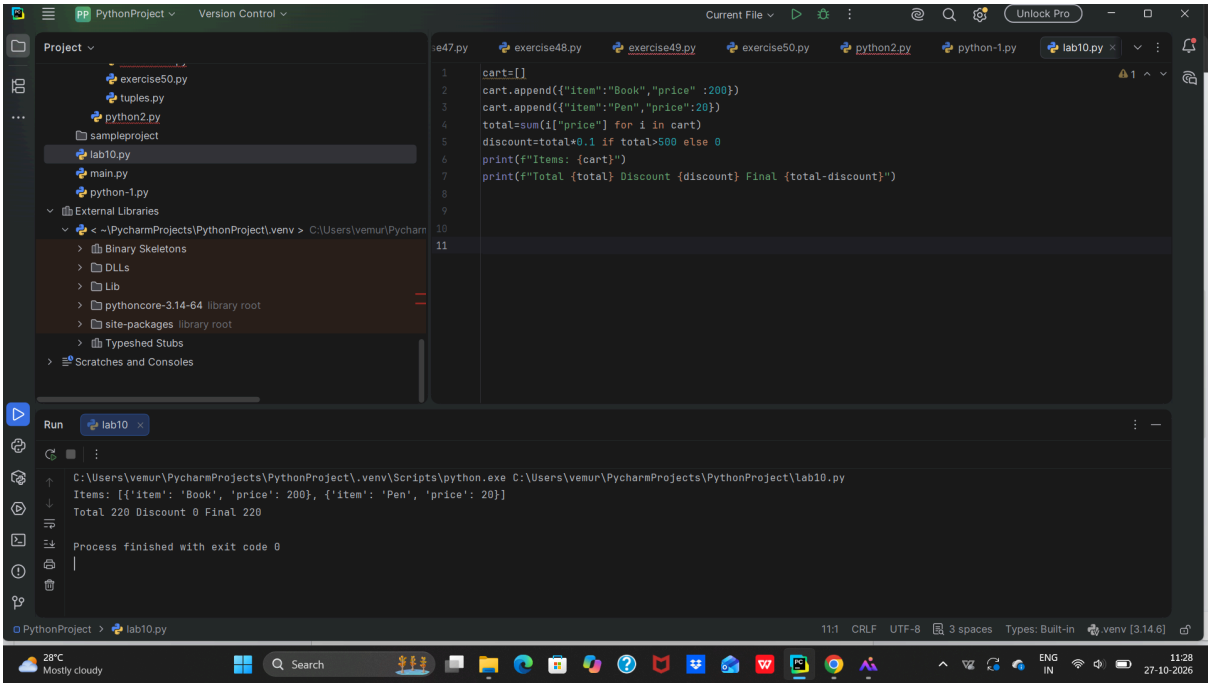

Code:

```
cart=[]
cart.append({"item":"Book","price" :200})
cart.append({"item":"Pen","price":20})
total=sum(i["price"] for i in cart)
discount=total*0.1 if total>500 else 0
print(f"Items: {cart}")
print(f"Total {total} Discount {discount} Final {total-discount}")
```

Output:

Items: [{'item': 'Book', 'price': 200}, {'item': 'Pen', 'price': 20}]
Total 220 Discount 0 Final 220

Screenshot:

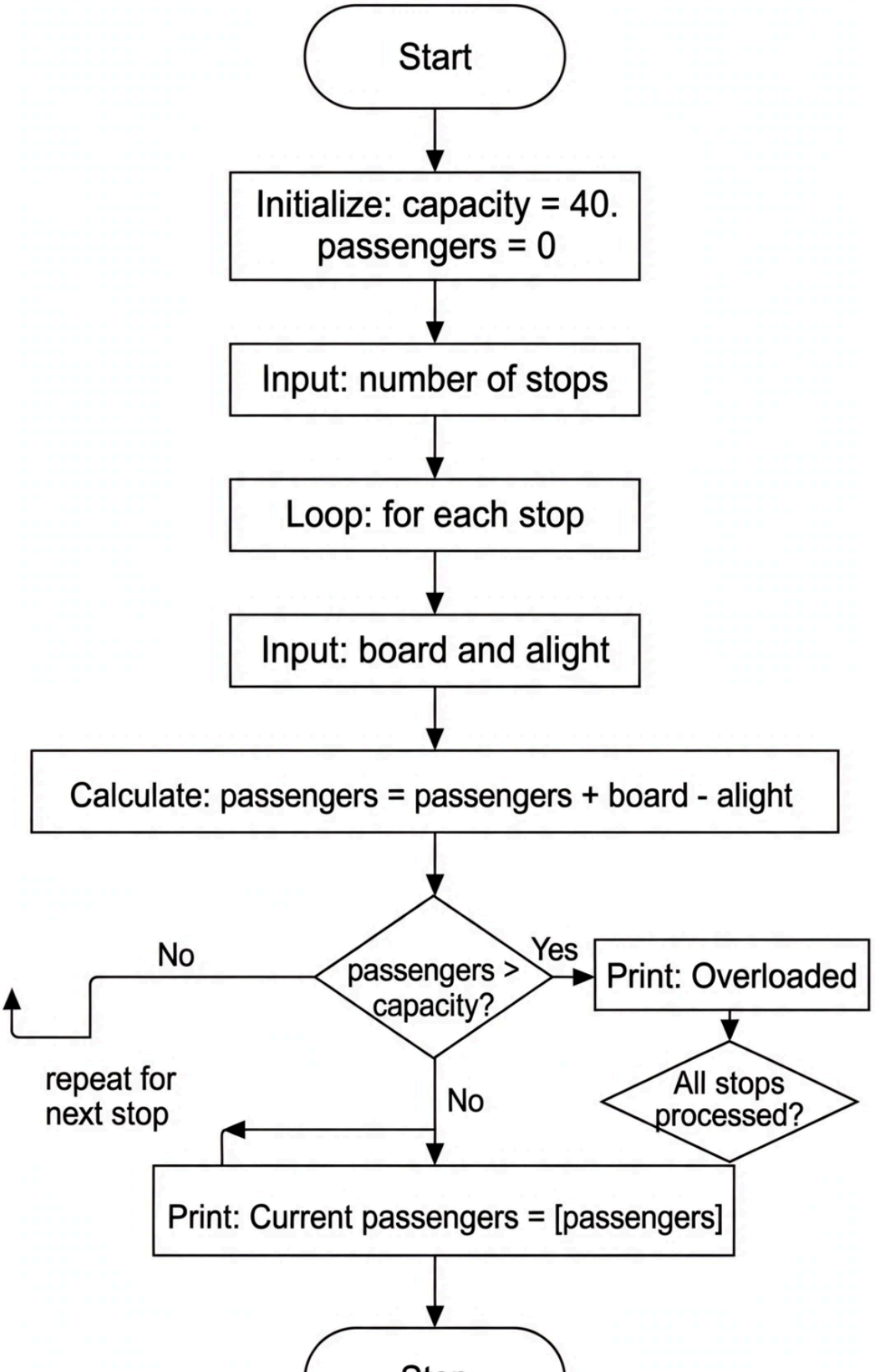


Problem Statement: Calculate Bus Passenger
Objective:We use if condition in this program.

Algorithm:
1.Start
2.capacity=40, passengers=0
3.Input stops
4.For each stop: input board, alight
5.passengers = passengers + board - alight
6.If passengers > capacity -> Overload else Current
7.Stop

Flowchart:

# Bus Passenger Occupancy Simulation



Code:

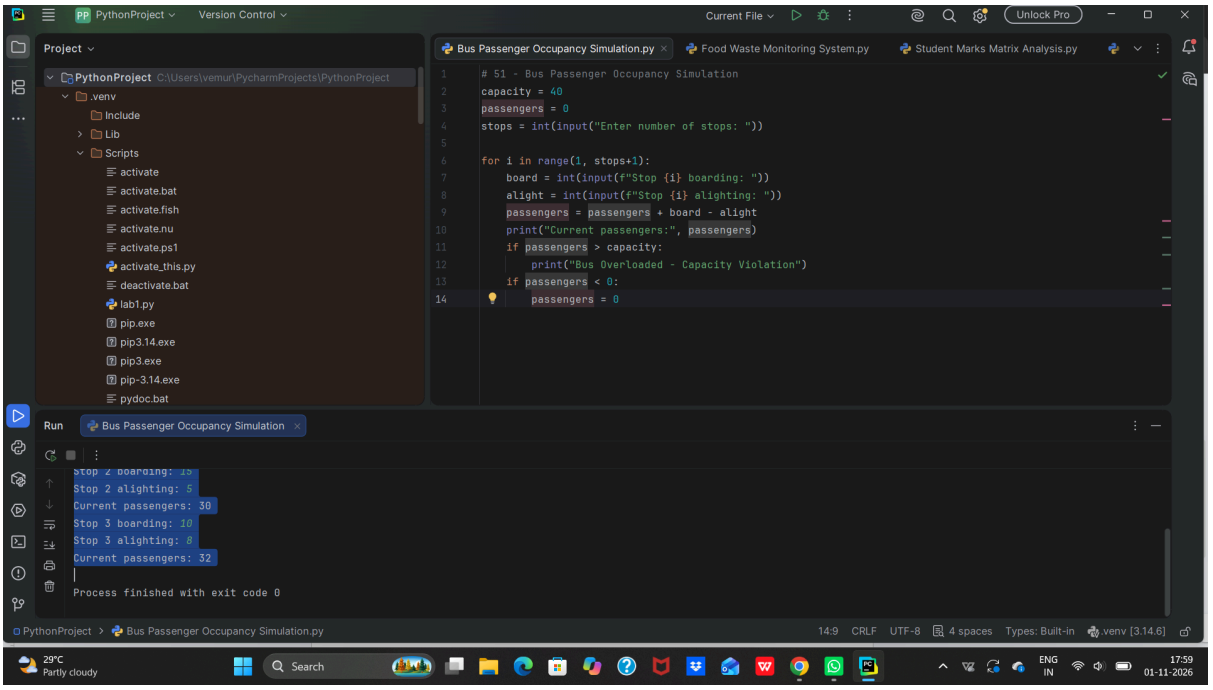
```
capacity = 40
passengers = 0
stops = int(input("Enter number of stops: "))

for i in range(1, stops+1):
   board = int(input(f"Stop {i} boarding: "))
   alight = int(input(f"Stop {i} alighting: "))
   passengers = passengers + board - alight
   print("Current passengers:", passengers)
   if passengers > capacity:
       print("Bus Overloaded - Capacity Violation")
   if passengers < 0:
       passengers = 0
```

Output:

Enter number of stops: 3
Stop 1 boarding: 20
Stop 1 alighting: 0
Current passengers: 20
Stop 2 boarding: 15
Stop 2 alighting: 5
Current passengers: 30
Stop 3 boarding: 10
Stop 3 alighting: 8
Current passengers: 32

Screenshot:



Problem Statement: Calculate Food Waste Monitoring System

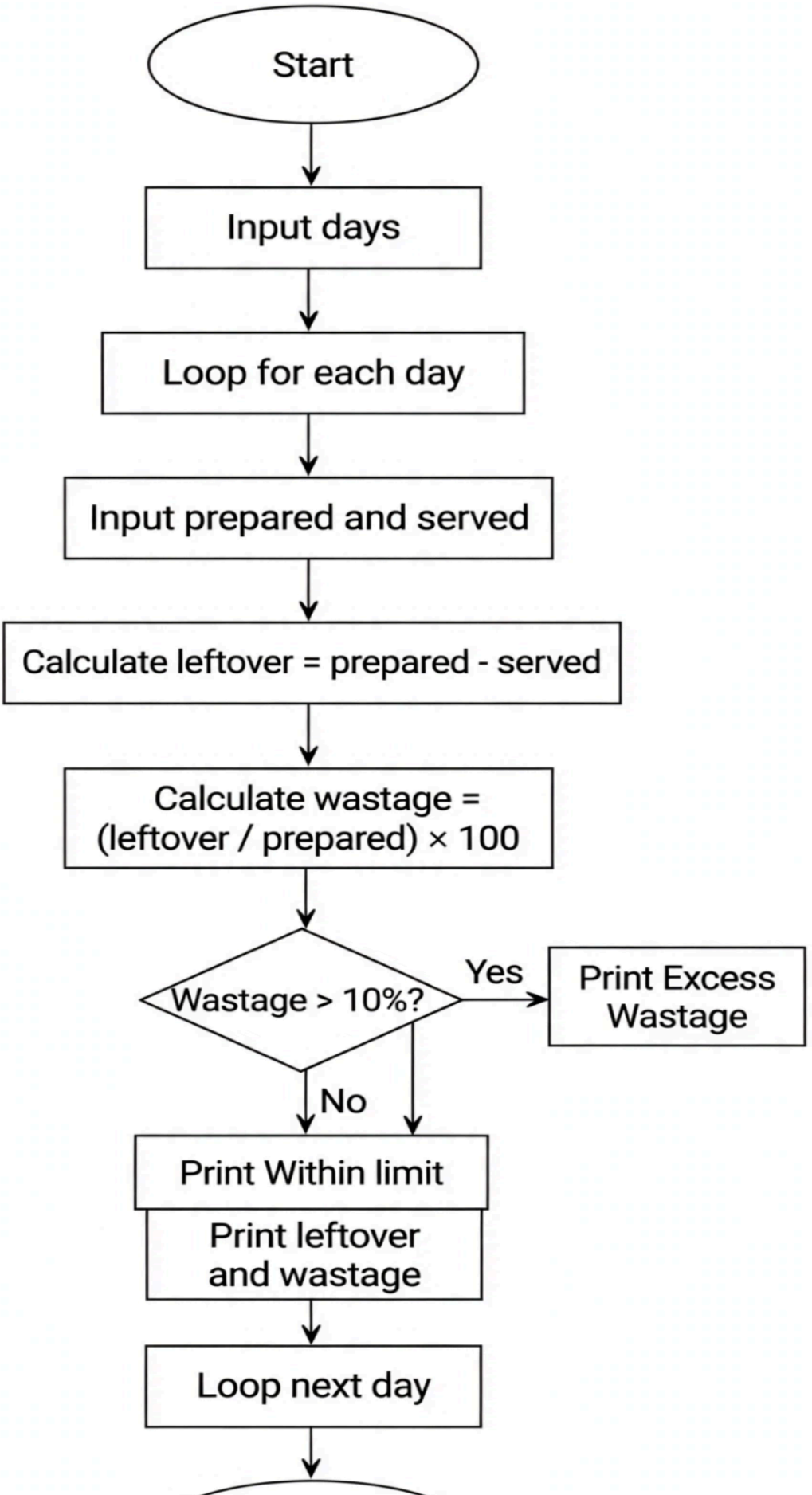
Objective:We use if-elif condition in this.

Algorithm:
1.Start
2.Input days
3.For each day: input prepared, served
4.leftover = prepared - served
5.wastage = (leftover/prepared)*100
6.If wastage > 10% -> Excess Wastage
7.Stop

Flowchart:

# Food Waste Monitoring System

Simple Flowchart



Code:

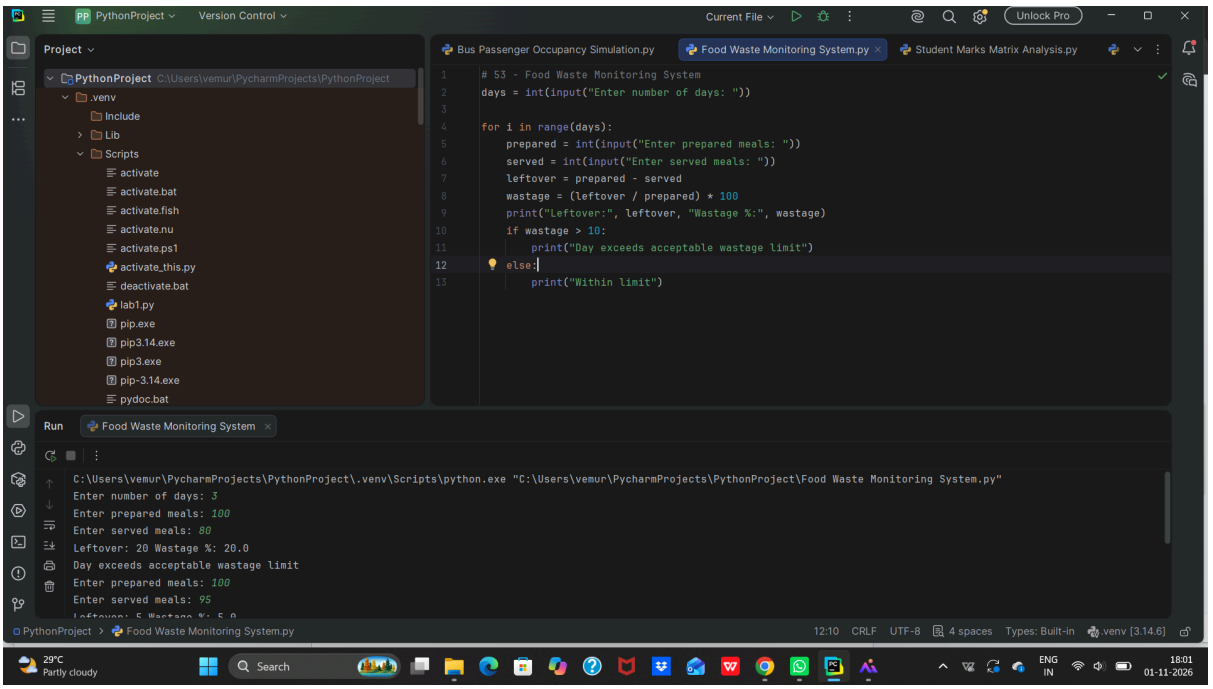
```
days = int(input("Enter number of days: "))

for i in range(days):
   prepared = int(input("Enter prepared meals: "))
   served = int(input("Enter served meals: "))
   leftover = prepared - served
   wastage = (leftover / prepared) * 100
   print("Leftover:", leftover, "Wastage %:", wastage)
   if wastage > 10:
       print("Day exceeds acceptable wastage limit")
   else:
       print("Within limit")
```

Output:

Enter number of days: 3
Enter prepared meals: 100
Enter served meals: 80
Leftover: 20 Wastage %: 20.0
Day exceeds acceptable wastage limit
Enter prepared meals: 100
Enter served meals: 95
Leftover: 5 Wastage %: 5.0
Within limit
Enter prepared meals: 50
Enter served meals: 30
Leftover: 20 Wastage %: 40.0
Day exceeds acceptable wastage limit

Screenshot:

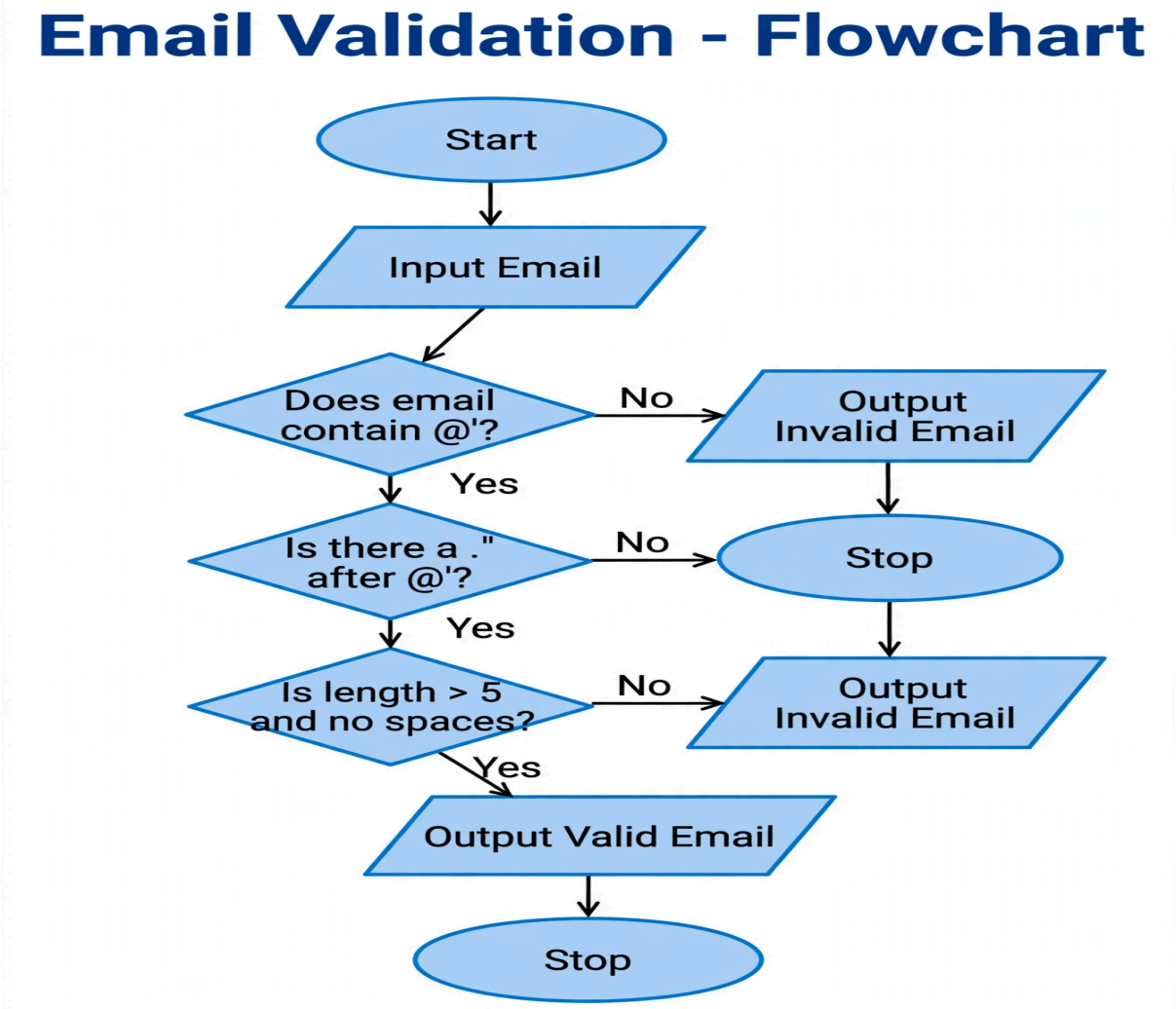

PythonProject
Version Control
Current File
Unlock Pro
Project
PythonProject C:\Users\vemur\PycharmProjects\PythonProject
.venv
Include
Lib
Scripts
activate
activate.bat
activate.fish
activate.nu
activate.ps1
activate_this.py
deactivate.bat
lab1.py
pip.exe
pip3.14.exe
pip3.exe
pip-3.14.exe
pydoc.bat
Bus Passenger Occupancy Simulation.py
Food Waste Monitoring System.py
Student Marks Matrix Analysis.py
# 53 - Food Waste Monitoring System
days = int(input("Enter number of days: "))
for i in range(days):
prepared = int(input("Enter prepared meals: "))
served = int(input("Enter served meals: "))
leftover = prepared - served
wastage = (leftover / prepared) * 100
print("Leftover:", leftover, "Wastage %:", wastage)
if wastage > 10:
print("Day exceeds acceptable wastage limit")
else:
print("Within limit")
Run
Food Waste Monitoring System
C:\Users\vemur\PycharmProjects\PythonProject\.venv\Scripts\python.exe "C:\Users\vemur\PycharmProjects\PythonProject\Food Waste Monitoring System.py"
Enter number of days: 3
Enter prepared meals: 100
Enter served meals: 80
Leftover: 20 Wastage %: 20.0
Day exceeds acceptable wastage limit
Enter prepared meals: 100
Enter served meals: 95
PythonProject
Food Waste Monitoring System.py
12:10
CRLF
UTF-8
4 spaces
Types: Built-in
venv [3.14.6]
29°C
Partly cloudy
Search
ENG
IN
18:01
01-11-2026

Problem Statement: Calculate email validation

Algorithm for email validation:

1.Start the program
2.Ask user for units
3.Check which slab the units fall
4.Calculate bill as per slab rate
5.Show bill amount
6.Stop

Flowchart:

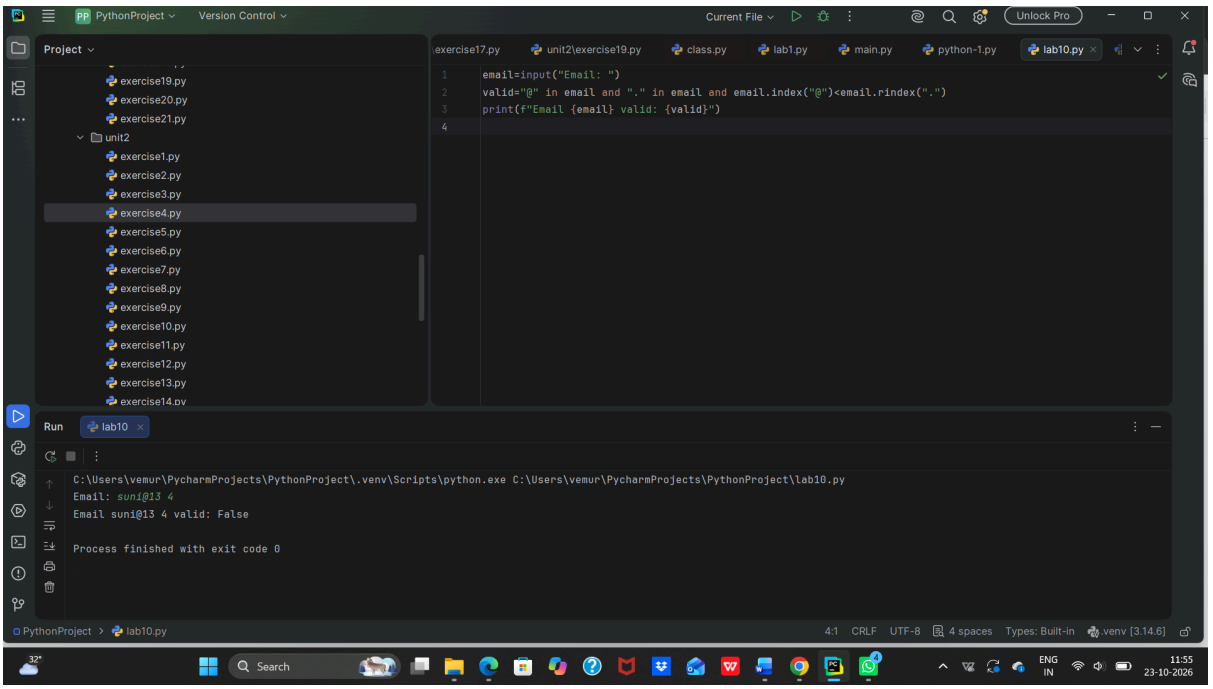



Code:

```
email=input("Email: ")
valid="@" in email and "." in email and email.index("@")<email.rindex(".")
print(f"Email {email} valid: {valid}")
```

Output:
Email: suni@13 4
Email suni@13 4 valid: False

Screenshot:

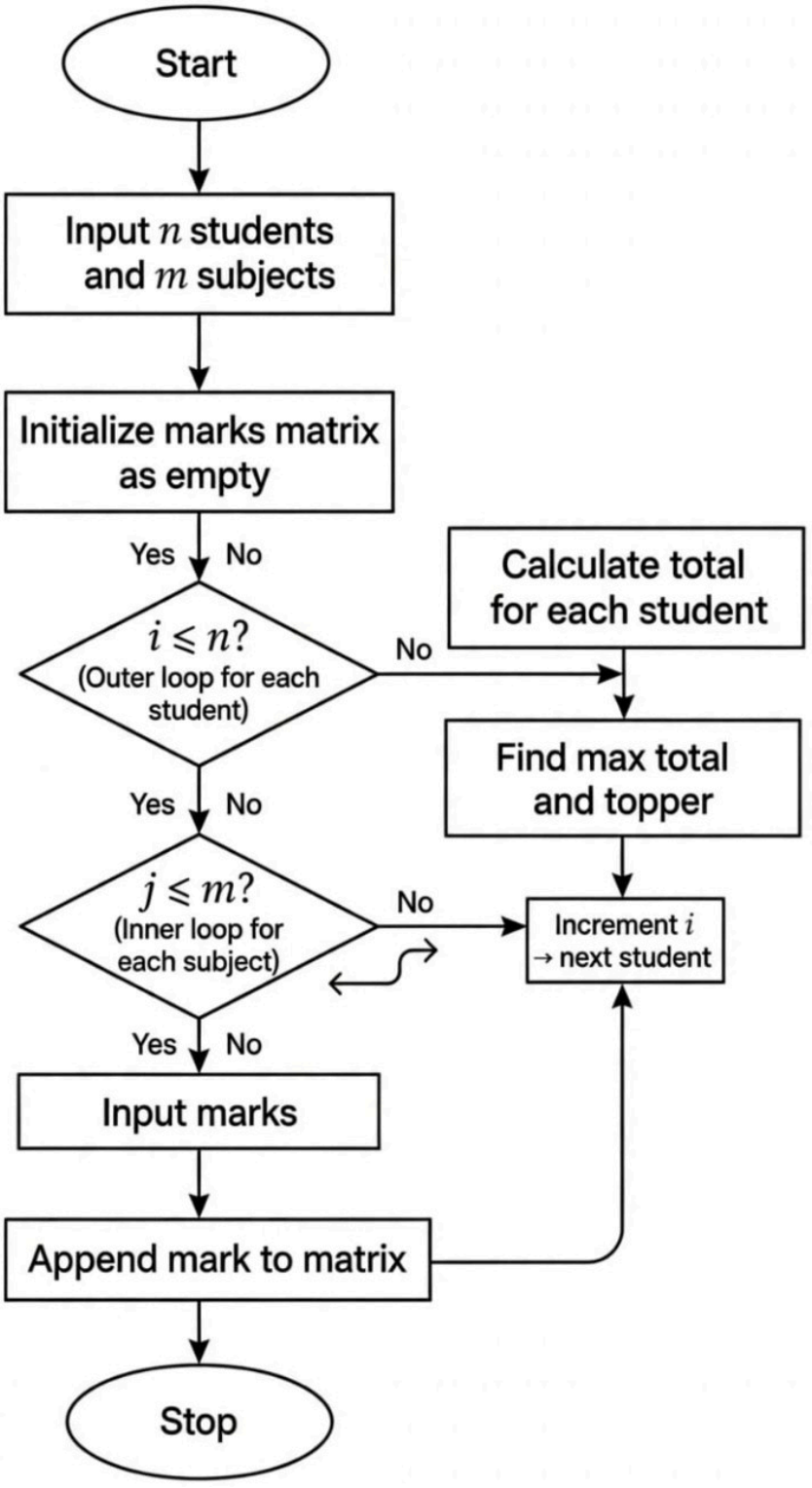

PythonProject
Version Control
Current File
Unlock Pro
Project
exercise19.py
exercise20.py
exercise21.py
unit2
exercise1.py
exercise2.py
exercise3.py
exercise4.py
exercise5.py
exercise6.py
exercise7.py
exercise8.py
exercise9.py
exercise10.py
exercise11.py
exercise12.py
exercise13.py
exercise14.py
exercise17.py
unit2\exercise19.py
class.py
lab1.py
main.py
python-1.py
lab10.py
email=input("Email: ")
valid="@" in email and "." in email and email.index("@")<email.rindex(".")
print(f"Email {email} valid: {valid}")
Run
lab10
C:\Users\vemur\PycharmProjects\PythonProject\.venv\Scripts\python.exe C:\Users\vemur\PycharmProjects\PythonProject\lab10.py
Email: suni@13 4
Email suni@13 4 valid: False
Process finished with exit code 0
PythonProject
lab10.py
4:1
CRLF
UTF-8
4 spaces
Types: Built-in
venv [3.14.6]
32°
Search
ENG
IN
11:55
23-10-2026

Problem Statement: Calculate Student Marks Matrix Analysis

Algorithm:
1.Start
2.Input n students, m subjects
3.Create marks matrix []
4.Outer loop for students, inner loop for subjects -> input marks
5.Calculate total for each student
6..Find max total -> Topper
7.Stop

Flowchart:

# Student Marks Matrix Analysis Flowchart



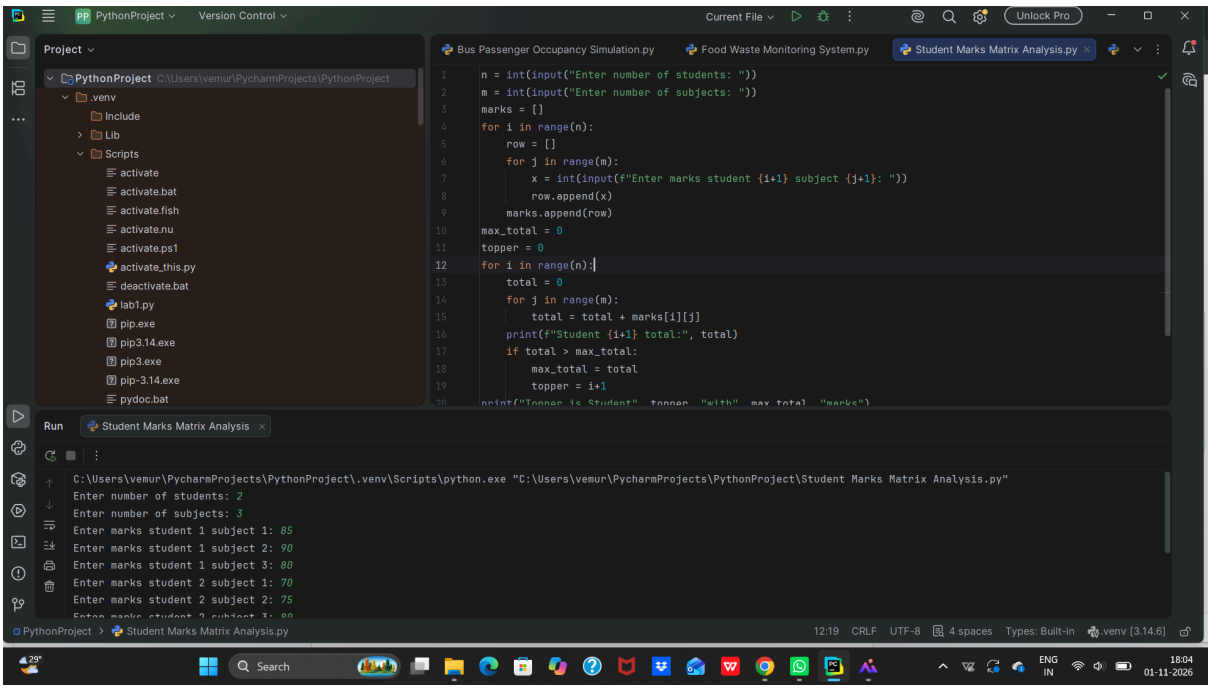
Code:

```
n = int(input("Enter number of students: "))
m = int(input("Enter number of subjects: "))
marks = []
for i in range(n):
    row = []
    for j in range(m):
        x = int(input(f"Enter marks student {i+1} subject {j+1}: "))
        row.append(x)
    marks.append(row)
max_total = 0
topper = 0
for i in range(n):
    total = 0
    for j in range(m):
        total = total + marks[i][j]
    print(f"Student {i+1} total:", total)
    if total > max_total:
        max_total = total
        topper = i+1
print("Topper is Student", topper, "with", max_total, "marks")
```

Output:

Enter number of students: 2
Enter number of subjects: 3
Enter marks student 1 subject 1: 85
Enter marks student 1 subject 2: 90
Enter marks student 1 subject 3: 80
Enter marks student 2 subject 1: 70
Enter marks student 2 subject 2: 75
Enter marks student 2 subject 3: 80
Student 1 total: 255
Student 2 total: 225
Topper is Student 1 with 255 marks

Screenshot:


PythonProject
Version Control
Current File
Unlock Pro
Project
PythonProject C:\Users\vemur\PycharmProjects\PythonProject
.venv
Include
Lib
Scripts
activate
activate.bat
activate.fish
activate.nu
activate.ps1
activate_this.py
deactivate.bat
lab1.py
pip.exe
pip3.14.exe
pip3.exe
pip-3.14.exe
pydoc.bat
Bus Passenger Occupancy Simulation.py
Food Waste Monitoring System.py
Student Marks Matrix Analysis.py
n = int(input("Enter number of students: "))
m = int(input("Enter number of subjects: "))
marks = []
for i in range(n):
row = []
for j in range(m):
x = int(input(f"Enter marks student {i+1} subject {j+1}: "))
row.append(x)
marks.append(row)
max_total = 0
topper = 0
for i in range(n):
total = 0
for j in range(m):
total = total + marks[i][j]
print(f"Student {i+1} total:", total)
if total > max_total:
max_total = total
topper = i+1
Run
Student Marks Matrix Analysis
C:\Users\vemur\PycharmProjects\PythonProject\.venv\Scripts\python.exe "C:\Users\vemur\PycharmProjects\PythonProject\Student Marks Matrix Analysis.py"
Enter number of students: 2
Enter number of subjects: 3
Enter marks student 1 subject 1: 85
Enter marks student 1 subject 2: 90
Enter marks student 1 subject 3: 80
Enter marks student 2 subject 1: 70
Enter marks student 2 subject 2: 75
PythonProject
Student Marks Matrix Analysis.py
12:19
CRLF
UTF-8
4 spaces
Types: Built-in
venv [3.14.6]
Search
ENG
IN
18:04
01-11-2026